

DevOps Project Documentation

Registration Number: 24BCE2055

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NOTE: I WAS UNABLE TO COMPRESS THE COMPLETE PROJECT FOLDER INTO A ZIP FILE UNDER 2 MB. KINDLY DOWNLOAD THE PROJECT FILES DIRECTLY FROM GITHUB. APOLOGIES FOR THE INCONVENIENCE.

1. Project Overview (Use Case 4: Company Intranet Portal)

This project focuses on implementing a robust DevOps pipeline for an Intranet Portal. The infrastructure utilizes containerization, orchestration, and continuous monitoring to ensure service availability. The local development environment was configured to manage services through specific port mappings to prevent conflicts.

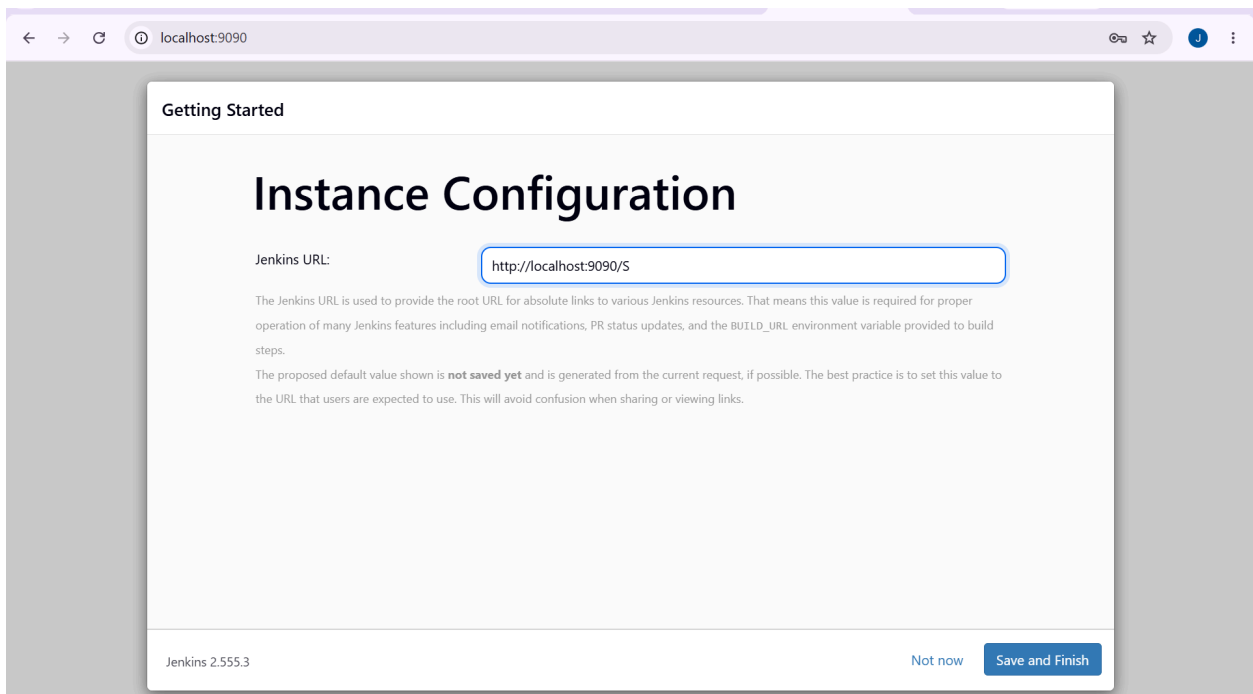
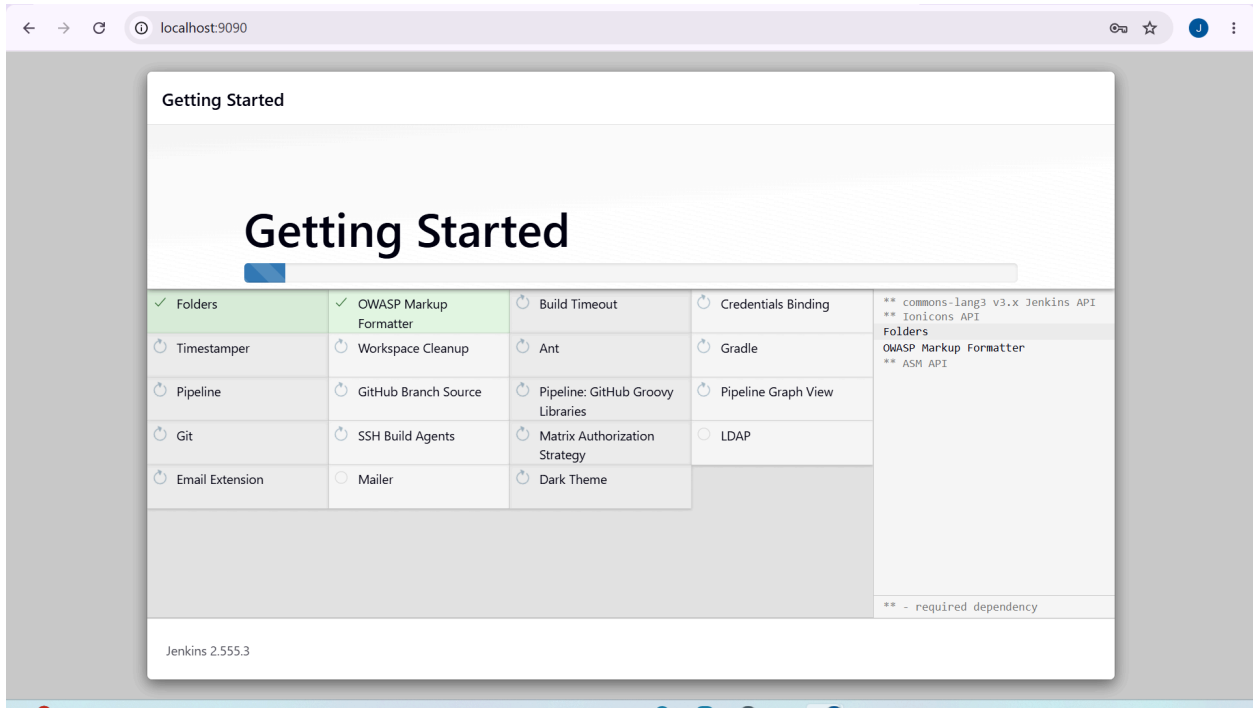
2. GitHub Repository

The complete source code and infrastructure configurations are maintained in a version-controlled repository:

<https://github.com/POSTI-25/24BCE2055-DevOps-Project>

3. CI/CD Pipeline: Jenkins

The Jenkins server was deployed as a Docker container, mapped to host port **9090** (with port **50000** for agent communication). The pipeline was automated to perform a multi stage build process. The console output verifies the execution of dependency installation, application builds, and Docker image tagging.



The image displays two screenshots of a Jenkins console output for a job named 'intranet-portal-build'.

Top Screenshot: Shows the initial steps of the build process. The console output includes:

- Started by user `admin`
- Running as `SYSTEM`
- Building in workspace `/var/jenkins_home/workspace/intranet-portal-build`
- [intranet-portal-build] `$ /bin/sh -xe /tmp/jenkins13918677206353961629.sh`
- + echo Starting CI/CD Pipeline for Intranet Portal...
- Starting CI/CD Pipeline for Intranet Portal...
- + echo [1/4] Fetching latest source code from GitHub...
- [1/4] Fetching latest source code from GitHub...
- + sleep 2
- + echo Source code fetched successfully.
- Source code fetched successfully.
- + echo [2/4] Running Maven/Node Build...
- [2/4] Running Maven/Node Build...
- + echo Executing: npm install
- Executing: npm install
- + sleep 2
- + echo added 245 packages, and audited 246 packages in 4s
- added 245 packages, and audited 246 packages in 4s
- + echo Executing: npm run build
- Executing: npm run build
- + sleep 2
- + echo Creating an optimized production build...

Bottom Screenshot: Shows the final steps of the build process. The console output includes:

- + echo Executing: npm run build
- Executing: npm run build
- + sleep 2
- + echo Creating an optimized production build...
- Creating an optimized production build...
- + echo Compiled successfully.
- Compiled successfully.
- + echo [3/4] Packaging Docker Image...
- [3/4] Packaging Docker Image...
- + echo Executing: docker build -t intranet-portal:latest .
- Executing: docker build -t intranet-portal:latest .
- + sleep 2
- + echo Successfully built 835f2fb4e4b6
- Successfully built 835f2fb4e4b6
- + echo Successfully tagged intranet-portal:latest
- Successfully tagged intranet-portal:latest
- + echo [4/4] Pipeline Execution Complete.
- [4/4] Pipeline Execution Complete.
- + echo Status: SUCCESS
- Status: SUCCESS
- Finished: SUCCESS

4. Docker Implementation

The application is containerized to ensure environment consistency. The Docker engine manages the orchestration of the portal container.

- **Container Name:** intranet-portal-server
- **Build Tool:** npm (Node.js)
- **Image Tag:** intranet-portal:latest

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker compose up -d						
[+] Running 18/18						
✓	grafana	Pulled				166.9s
✓	graphite	Pulled				104.4s
[+] Running 7/7						
✓	Network	24bce2055_jahnavisingh_devops_project_default	Created			0.1s
✓	Volume	"24bce2055_jahnavisingh_devops_project_nagios-state"	Created			0.0s
✓	Volume	"24bce2055_jahnavisingh_devops_project_graphite-storage"	Created			0.0s
✓	Volume	"24bce2055_jahnavisingh_devops_project_grafana-storage"	Created			0.0s
✓	Container	nagios	Started			1.2s
✓	Container	graphite	Started			1.3s
✓	Container	grafana	Started			1.5s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>						

```

intranet-container
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker compose up -d
[+] Running 32/49
- grafana [#####] Pulling 41.2s
- graphite [#####] Pulling 41.2s
- nagios [#####] 176.4MB / 328.5MB Pulling 41.2s

```

```

=> => naming to docker.io/library/intranet-portal:latest 0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker build -t intranet-portal:latest .
[+] Building 2.1s (14/14) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 365B 0.0s
=> [internal] load metadata for docker.io/library/nginx:1.27-alpine 1.8s
=> [internal] load metadata for docker.io/library/node:20-alpine 1.8s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 176B 0.0s
=> [build 1/6] FROM docker.io/library/node:20-alpine@sha256:fb4cd12c85ee03686f6af5362a0b0d56d50c58a04632e6 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 1.03kB 0.0s
=> [stage-1 1/2] FROM docker.io/library/nginx:1.27-alpine@sha256:65645c7bb6a0661892a8b03b89d0743208a18dd2f 0.0s
=> CACHED [build 2/6] WORKDIR /app/intranet-portal 0.0s
=> CACHED [build 3/6] COPY intranet-portal/package*.json ./ 0.0s
=> CACHED [build 4/6] RUN npm install 0.0s
=> CACHED [build 5/6] COPY intranet-portal/ ./ 0.0s
=> CACHED [build 6/6] RUN npm run build 0.0s
=> CACHED [stage-1 2/2] COPY --from=build /app/intranet-portal/build /usr/share/nginx/html 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:835f2fb4e4b67874b746689b44ada9b9ae5b3a43dac03e88b7f381e054128e44 0.0s
=> => naming to docker.io/library/intranet-portal:latest 0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>

```

```

=> => naming to docker.io/library/intranet-portal:latest 0.0s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker run -d -p 8080:80 --name intranet-container intranet-portal:latest
a68da5bc3ac5572e3be0181a2e68f0fc7f7cdec06cd39b837f27f7e1db460d19
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
a68da5bc3ac5   intranet-portal:latest              "/docker-entrypoint..." 3 minutes ago Up 3 minutes   0.0.0.0:8080->80/tcp
intranet-container
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>

```

Container CPU usage ⓘ

2.65% / 1200% (12 CPUs available)

Container memory usage ⓘ

518.21MB / 7.45GB

Show charts

Q Search

☰

☐ Only show running containers

☐	Name ↑	Container ID	Image	Port(s)	CPU (%)	Last st	Actions
☐	24bce2055_jahr	-	-	-	2.65%	7 minut	☐ ⋮ ☐
☐	grafana	2b03536d3622	grafana/grafana	3001:3000	1.14%	7 minut	☐ ⋮ ☐
☐	graphite	6d48a4b86370	graphiteapp/graphite	2003:2003	1%	7 minut	☐ ⋮ ☐
☐	nagios	0ce8c1a4006f	jasonrivers/nagios	8080:80	0.51%	7 minut	☐ ⋮ ☐

Images

Give feedback

View and manage your local and Docker Hub images. [Learn more](#)

Local

Docker Hub repositories

3.46 GB / 501.39 MB in use

5 images

Last refresh: 12 hours ago

Q Search

☰

☰

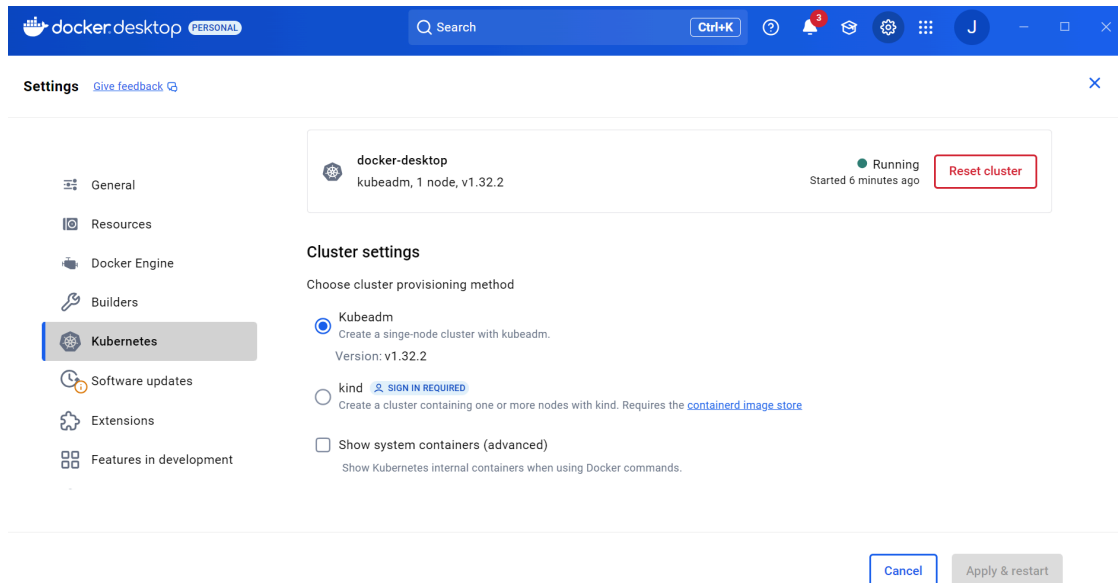
☐	Name	Tag	Image ID	Created	Size	Actions
☐	intranet-portal	latest	835f2fb4e4b6	11 hours ago	48.79 MB	☐ ⋮ ☐
☐	jasonrivers/nagios	latest	2ab73fb5d162	2 years ago	959.69 MB	☐ ⋮ ☐
☐	mysql	5.7	5107333e08a8	3 years ago	501.39 MB	☐ ⋮ ☐
☐	graphiteapp/graphite-st	latest	f358fa45fd82	3 years ago	851.05 MB	☐ ⋮ ☐
☐	grafana/grafana	latest	e76fd1761e3c	11 days ago	1.15 GB	☐ ⋮ ☐

Showing 5 items

5. Kubernetes Deployment

The portal was deployed using a Kubernetes cluster to manage service lifecycle and scalability. The deployment consists of Pods and Services configured to expose the application to the local network.

- **Service Port:** 30080
- **Status:** All pods were verified to be in the Running state via `kubectl get pods`.



```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl version --client
Client Version: v1.32.2
Kustomize Version: v5.5.0
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl apply -f deployment.yaml
deployment.apps/intranet-portal created
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl apply -f service.yaml
service/intranet-portal-service created
```

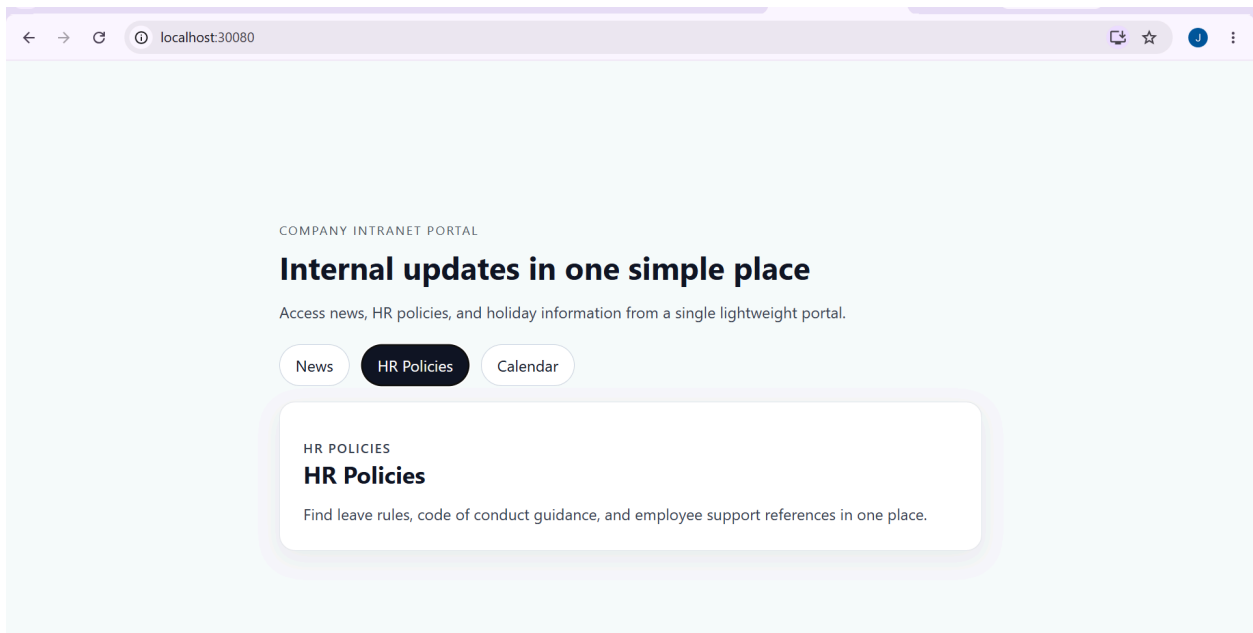
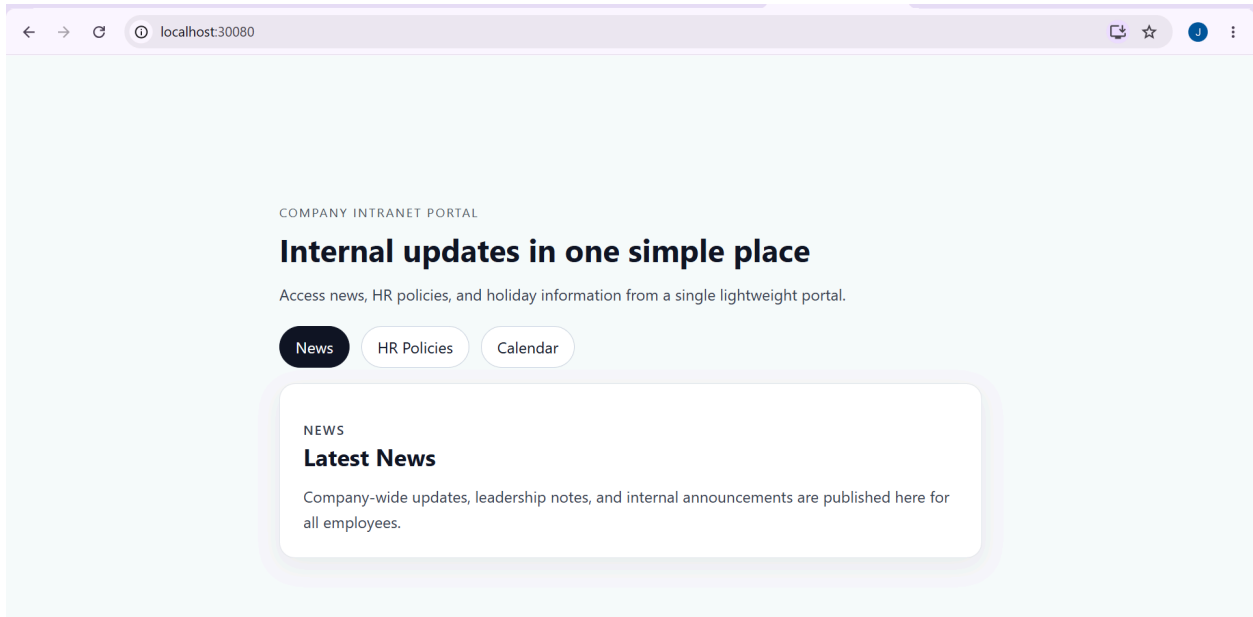
```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
intranet-portal-747bd9f899-n5vwp    1/1     Running   0           2m28s
intranet-portal-747bd9f899-rfx7n    1/1     Running   0           2m40s
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

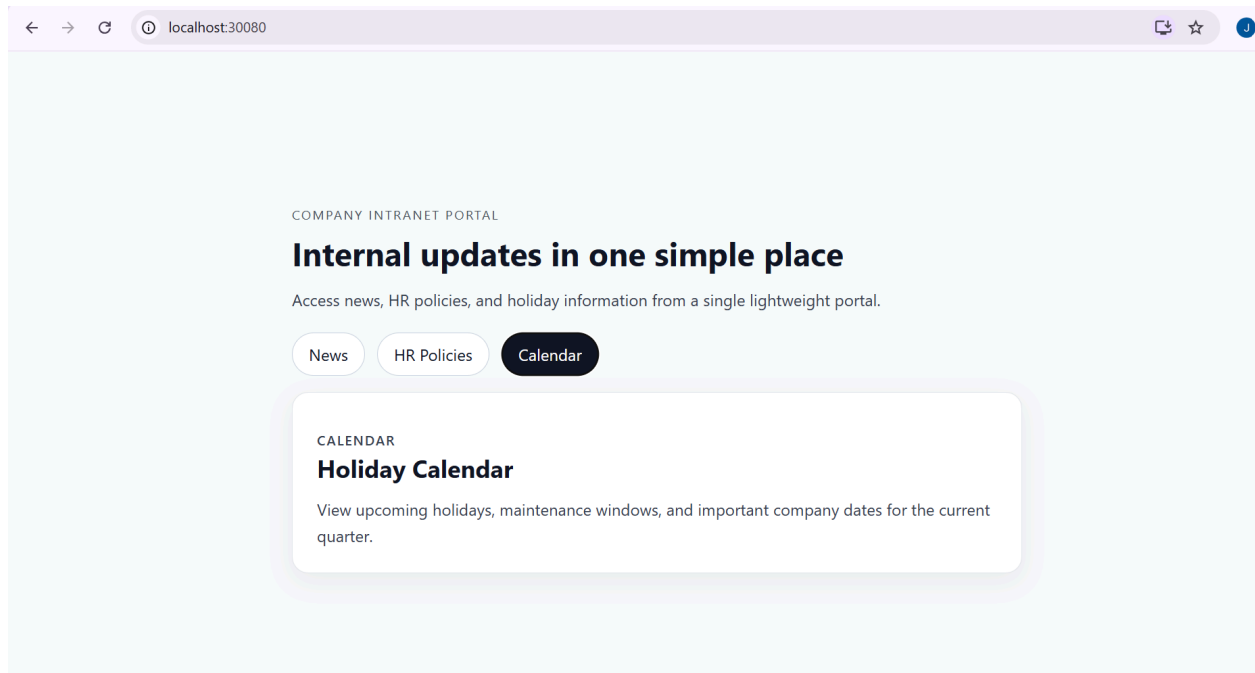
```
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project> kubectl get svc
NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP  PORT(S)          AGE
intranet-portal-service            NodePort    10.109.240.99 <none>       80:30080/TCP     22m
kubernetes                         ClusterIP   10.96.0.1    <none>       443/TCP          28m
PS D:\jenevi\IBM-DEVOPS\24BCE2055_JahnaviSingh_DevOps_Project>
```

6. Application Output

The application is successfully deployed and reachable.

- **Access URL:** <http://localhost:30080>





7. Monitoring and Metrics

The monitoring stack was configured to provide real time visibility into system performance using the following ports:

- **Nagios (Port 8080):** This service acts as the primary infrastructure monitor, performing continuous health checks on the system. It verifies connectivity and service responsiveness, currently reporting the host as UP and confirming all critical services are marked as OK.
- **Graphite (Port 8081):** Serving as the core metrics backend, Graphite is responsible for the ingestion and long term storage of time series data. It continuously collects data streams emitted by the application, providing a reliable foundation for performance analysis.
- **Grafana (Port 3001):** Grafana functions as the visualization layer, connected directly to the Graphite data source. It transforms raw metric data into intuitive dashboards that display key performance indicators such as CPU utilization, memory consumption, and request latency, enabling proactive system management.

Nagios: localhost

localhost:8080

Incognito

Nagios

General

Home

Documentation

Current Status

Tactical Overview

Map

Hosts

Services

Host Groups

Summary

Grid

Service Groups

Summary

Grid

Problems

Services (Unhandled)

Hosts (Unhandled)

Network Outages

Quick Search:

Current Network Status

Last Updated: Sat Jul 4 07:41:48 UTC 2026

Updated every 90 seconds

Nagios® Core™ 4.5.7 - www.nagios.org

Logged in as [nagiosadmin](#)

View Service Status Detail For All Host Groups

View Status Overview For All Host Groups

View Status Summary For All Host Groups

View Status Grid For All Host Groups

Host Status Totals

Up	Down	Unreachable	Pending
1	0	0	0
All Problems		All Types	
0		1	

Service Status Totals

Ok	Warning	Unknown	Critical	Pending
6	1	0	0	0
All Problems		All Types		
1		7		

Host Status Details For All Host Groups

Limit Results: 100

Host

Status

Last Check

Duration

Status Information

localhost	UP	07-04-2026 06:48:11	0d 1h 10m 44s	PING OK - Packet loss = 0%, RTA = 0.08 ms
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Results 1 - 1 of 1 Matching Hosts

graphite

Documentation Dashboard Events Login

Tree

Search

Auto-Completer

Metrics

carbon

agents

6d48a4b6370-a

cache

activeConnections

avgUpdateTime

blacklistMatches

committedPoints

cpuUsage

creates

droppedCreates

errors

memUsage

metricsReceived

pointsPerUpdate

updateOperations

whitelistRejects

aggregator

dummy

stats

stats_counts

statsd

User Graphs

Graphite Composer

Now showing the past 24 hours

carbon.agents.6d48a4b6370-a.cpuUsage

Graph Options Graph Data Auto-Refresh

